

Course- BTech
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Type- Core
Course Name- Computational Thinking and Programming
Semester- odd
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Tutorial Assignment: 6

Tutorial title: Sequence Operations: Tuple, List, dictionary, strings

CO Mapping

Name	CO1	CO2	CO3
Sequence Operations	✓	-	-

CONCEPT CLARITY:

Sequence Data Types are used to store data in containers in the Python. The different types of containers used to store the data are **List, Tuple, Set, Dictionary** and **String**.

- **Mutable Data Types:** Data types in python where the value assigned to a variable can be changed
- **Immutable Data Types:** Data types in python where the value assigned to a variable cannot be changed

Data Structure	Ordered	Mutable	Constructor	Example
List	Yes	Yes	[] or list()	[5.7, 4, 'yes', 5.7]
Tuple	Yes	No	() or tuple()	(5.7, 4, 'yes', 5.7)
Set	No	Yes	{ }* or set()	{5.7, 4, 'yes'}
Dictionary	No	Yes**	{ } or dict()	{'Jun': 75, 'Jul': 89}

Sequence Operators

Concatenation : +

Repetition : *

Membership Testing : x in S

Slicing a Sequence : S[i:j], S[i:], S[:j]

List Methods

Method	Description
<code>L.count(x)</code>	Returns the number of items of <code>L</code> that are equal to <code>x</code> .
<code>L.index(x)</code>	Returns the index of the first occurrence of an item in <code>L</code> that is equal to <code>x</code> , or raises an exception if <code>L</code> has no such item.
<code>L.append(x)</code>	Appends item <code>x</code> to the end of <code>L</code> ; e.g., <code>L[len(L):]=[x]</code> .
<code>L.extend(s)</code>	Appends all the items of iterable <code>s</code> to the end of <code>L</code> ; e.g., <code>L[len(L):]=s</code> .
<code>L.insert(i, x)</code>	Inserts item <code>x</code> in <code>L</code> before the item at index <code>i</code> , moving following items of <code>L</code> (if any) "rightward" to make space (increases <code>len(L)</code> by one, does not replace any item, does not raise exceptions: acts just like <code>L[i:i]=[x]</code>).
<code>L.remove(x)</code>	Removes from <code>L</code> the first occurrence of an item in <code>L</code> that is equal to <code>x</code> , or raises an exception if <code>L</code> has no such item.
<code>L.pop([i])</code>	Returns the value of the item at index <code>i</code> and removes it from <code>L</code> ; if <code>i</code> is omitted, removes and returns the last item; raises an exception if <code>L</code> is empty or <code>i</code> is an invalid index in <code>L</code> .
<code>L.reverse()</code>	Reverses, in place, the items of <code>L</code> .
<code>L.sort([f])</code> (2.3)	Sorts, in place, the items of <code>L</code> , comparing items pairwise via function <code>f</code> ; if <code>f</code> is omitted, comparison is via the built-in function <code>cmp</code> . For more details, see Sorting a list .
<code>L.sort(cmp=cmp, key=None, reverse=False)</code> (2.4)	Sorts, in-place, the items of <code>L</code> , comparing items pairwise via the function passed as <code>cmp</code> (by default, the built-in function <code>cmp</code>). When argument <code>key</code> is not <code>None</code> , what gets compared for each item <code>x</code> is <code>key(x)</code> , not <code>x</code> itself. For more details, see Sorting a list .

EXERCISE

Q1: Create a list containing five different types of data elements (e.g., integer, string, float, list, tuple).

Q2: Given a nested list `[[1, 2, 3], [4, 5, 6], [7, 8, 9]]`, how would you access the number 5?

Q3: Write a loop to print each element of the list `['apple', 'banana', 'cherry']`. How would you add the elements 'grape' and 'orange' to the end of the list `['apple', 'banana', 'cherry']`?

Q4: Given the list [0, 1, 2, 3, 4, 5], retrieve a sublist containing just [2, 3, 4].

Q5: What method would you use to remove the first occurrence of the value 3 from the list [1, 2, 3, 4, 3, 5]?

Q6: Use list comprehension to create a list of squares for numbers from 1 to 10.

Q7: Use list comprehension to create a list of even numbers from 1 to 20.

Q8: Given a matrix [[1, 2, 3], [4, 5, 6], [7, 8, 9]], use list comprehension to flatten it into a single list.

Q9: Check if the value 5 is present in the list [1, 2, 3, 4, 6, 7].

Q10: Write a loop to iterate over the list ['a', 'b', 'c', 'd'] and print both the index and value of each element.

Q11: Use list comprehension to create a list of vowels present in the word 'elephant'.

Q10: How would you count the number of occurrences of the value 2 in the list [1, 2, 3, 2, 4, 2, 5]?

Q11: Use list comprehension to generate all possible pairs of elements from the lists ['a', 'b'] and [1, 2, 3].

Q12: Given a nested list [[1, 2], [3, 4], [5, 6]], how would you check if the value 4 is present in one of the inner lists?

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Bonus question:

Sita is working in a company in IT department. She has been given task to check whether last password used by employees in the entry card is not same as new password. She does not want last password and new password to have same characters but varying order of occurrence. She also wants length of password to be same as old password.